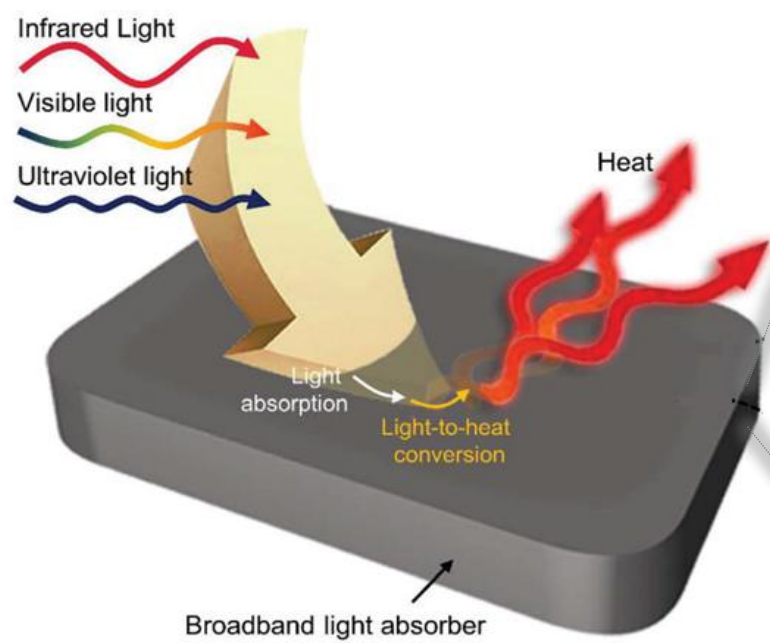


그림 1. 태양광 에너지를 이용한 수증기 발생 원리

Photothermal conversion



Solar vapor generation system

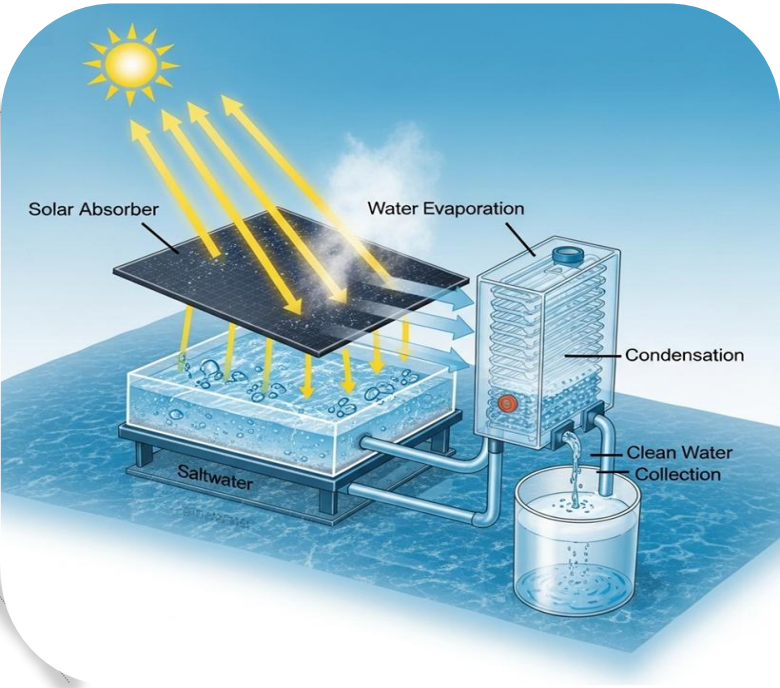


그림 2. 블랙 TiO₂@C 합성 과정

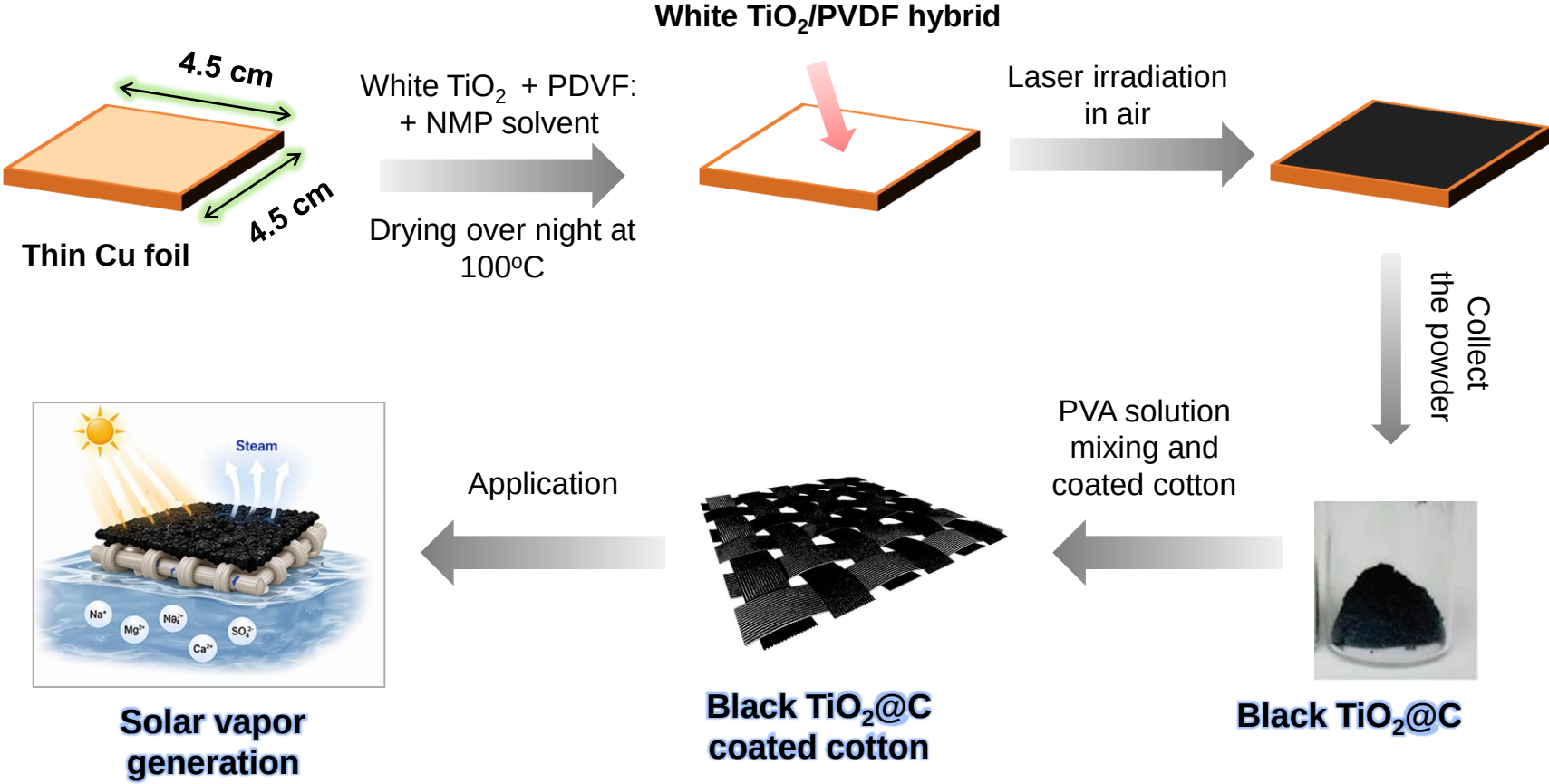


그림 3.블랙 TiO2@C(Laser Scribed TiO2 with PVDF), Raw TiO2 (W-TiO2), Laser Scribed TiO2 without PVDF (L-TiO2)의 형상 비교

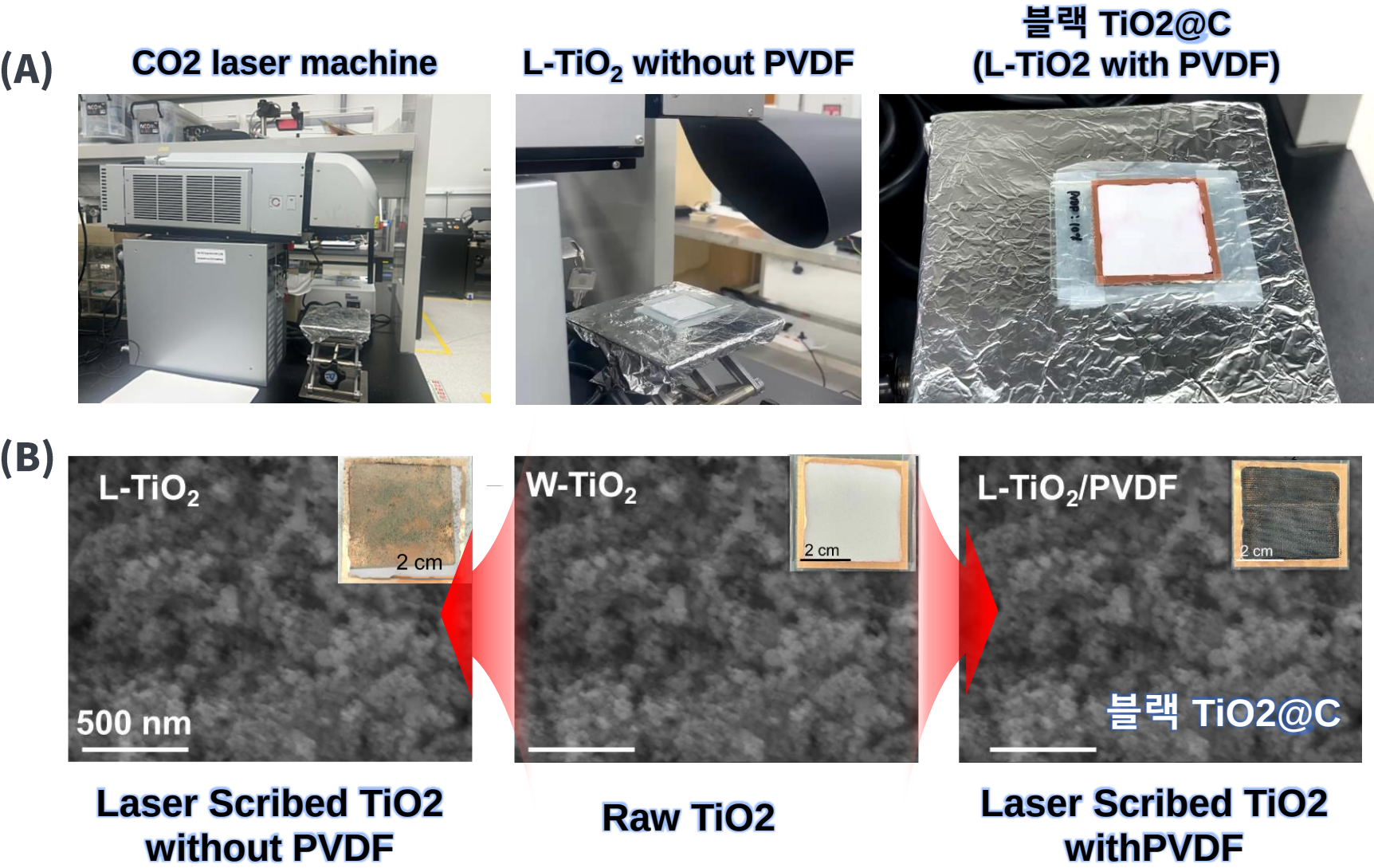


그림 4.블랙 TiO₂@C(Laser Scribed TiO₂ with PVDF), Raw TiO₂ (W-TiO₂), Laser Scribed TiO₂ without PVDF (L-TiO₂)의 XRD, Raman spectroscopy, TEM, XPS 분석 결과

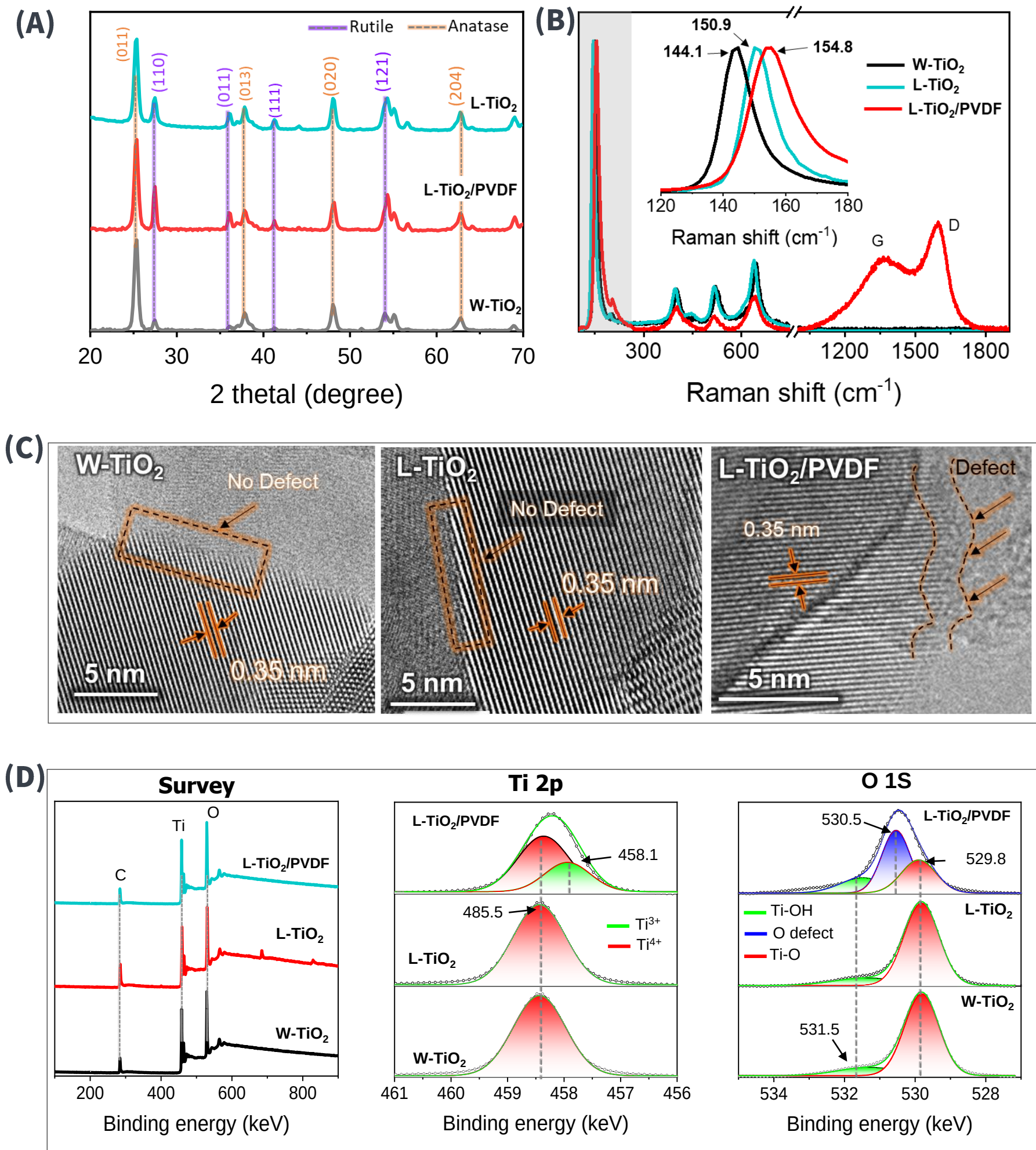
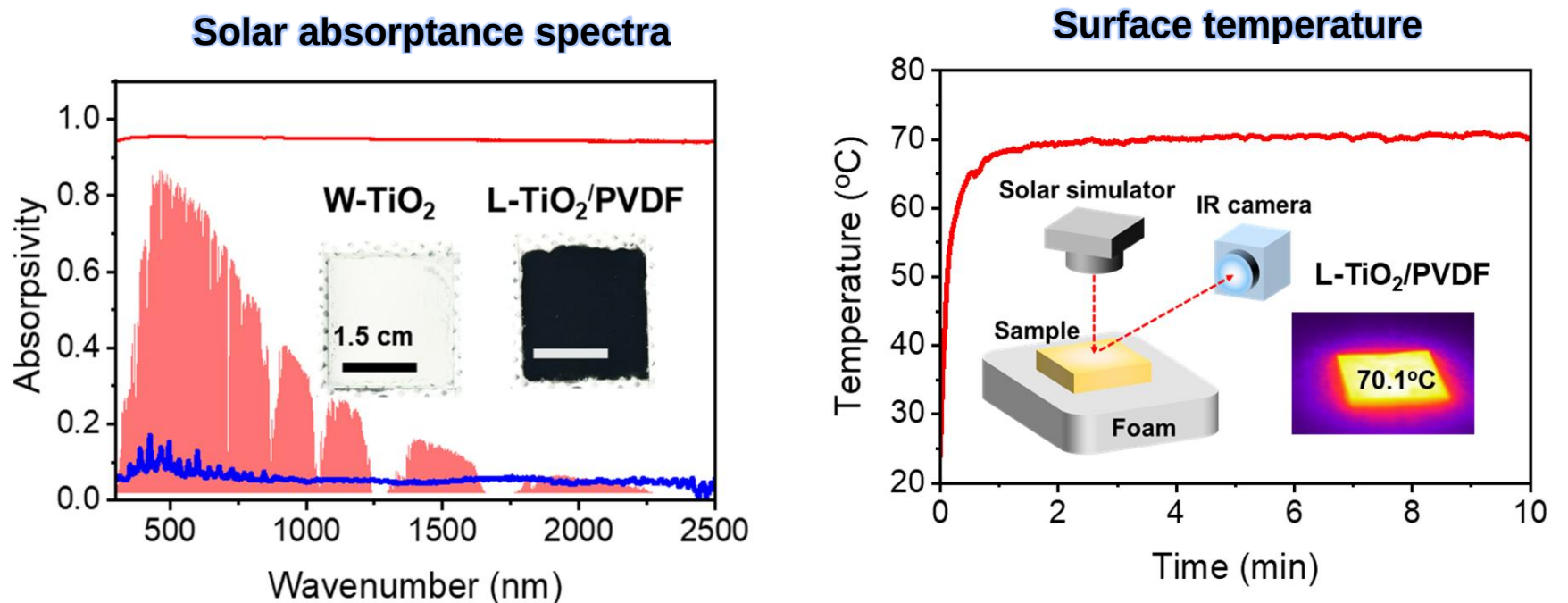


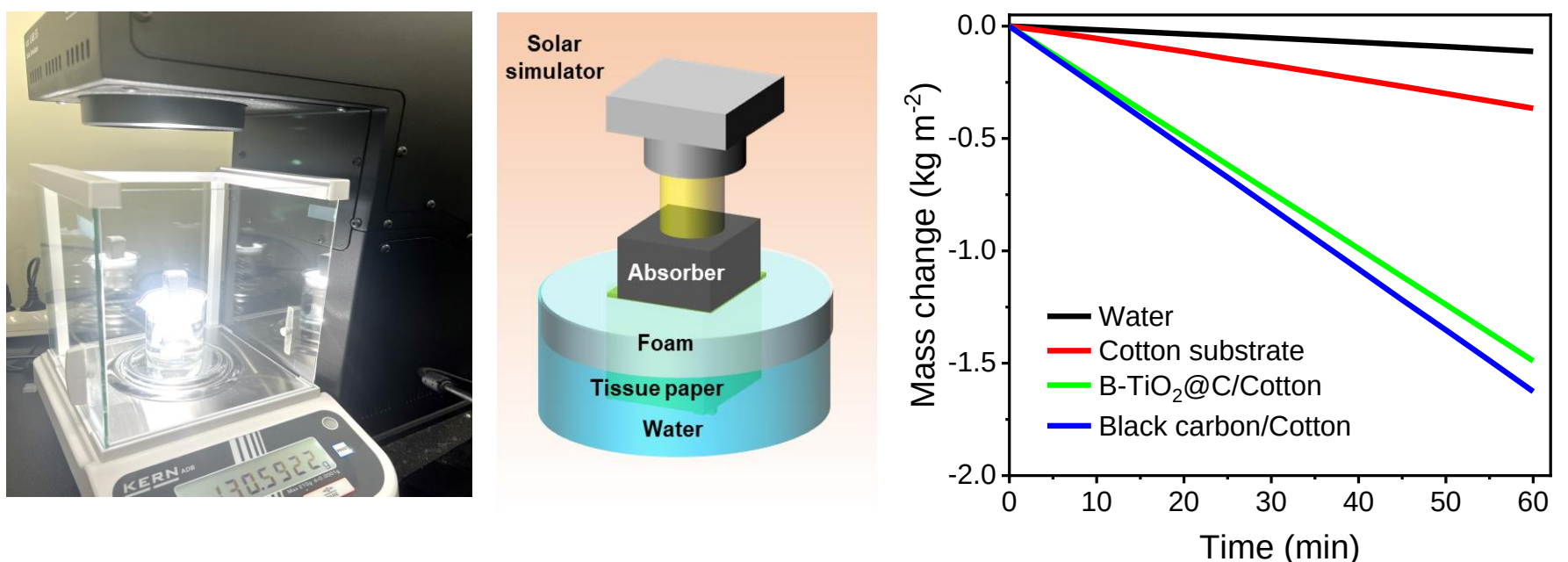
그림 5.블랙 TiO₂@C(Laser Scribed TiO₂ with PVDF), Raw TiO₂ (W-TiO₂)의 광학적 특성 및 광열 변환 성능 비교

(A)



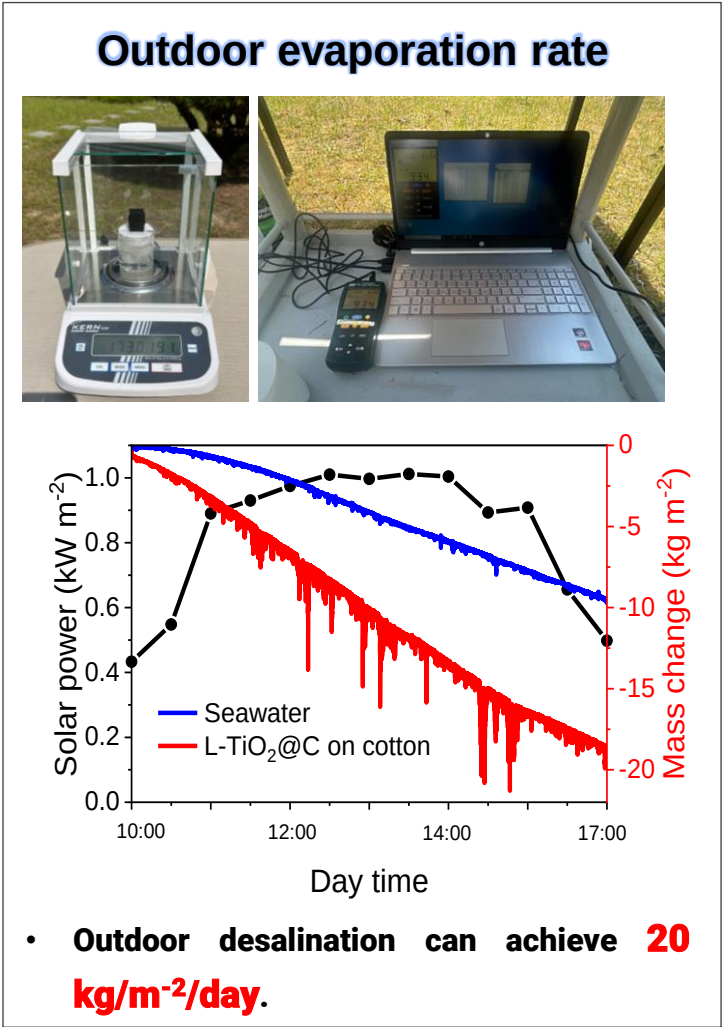
- The sample exhibits high solar absorptance (**~94%**).
- Surface temperature reaches **~70.1°C** under 1 sun illumination.

(B)

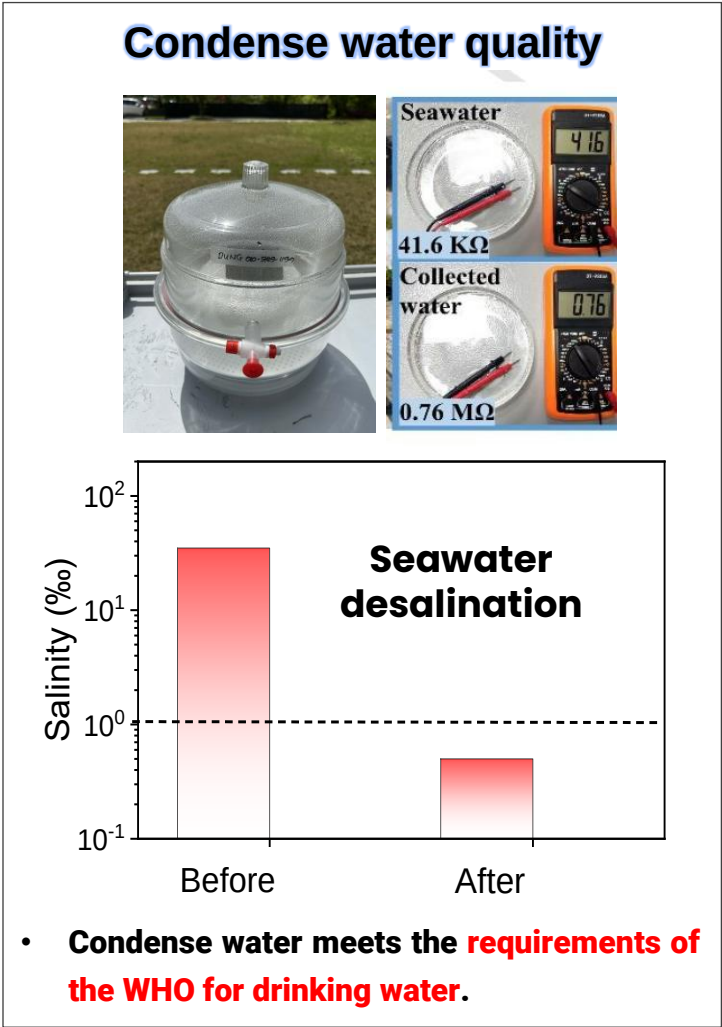


- B-TiO₂@C coated cotton has a water evaporation rate of **1.51 kg m⁻²h⁻¹**

그림 6. 블랙 TiO₂@C(Laser Scribed TiO₂ with PVDF), 증발 속도 및 야외 담수화 결과



(A)



(B)